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February 11, 1975
SUPERSEDING
Fed. Spec. GGG-C-406c
April 5, 1967

FEDERAL SPECIFICATION

CLAMPS AND HAND SCREWS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers clamps and hand screws used for clamping and holding work in a fixed position.

1.1.1 Federal specification coverage. This specification does not cover all varieties of clamps or hand screws that are commercially available, but is intended to cover only those varieties generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, styles, and sizes. The clamps and hand screws shall be of the following types, classes, and styles, as specified (see 6.1):

Type I - Clamp "C".
  Class E - Very light service.
    Style 1 - Regular throat.
    Style 2 - Deep throat.
  Class A - Light service.
    Style 1 - Regular throat.
    Style 2 - Deep throat.
  Class B - Medium service.
    Style 1 - Regular throat.
    Style 2 - Deep throat.
  Class D - Heavy service.

Type III - Clamp, screw, quick-adjusting.
  Class A - Fixed head.
  Class B - With sliding head.
  Class C - Threaded head.

Type IV - Hand screw.

Type V - Clamp, cantilever jaw.

1.2.1.1 Size. The size of the clamp or hand screw shall be as shown in the applicable table and as specified (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

FSC 5120

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is specified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standard for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive, not restrictive, and are not intended to preclude the purchase of clamps otherwise conforming to this specification.

3.2 Materials. Materials used in the manufacture of clamps and hand screws shall be of uniform quality and free from defects and imperfections that may affect their serviceability or durability.

3.2.1 Screws. The screws shall be of steel and shall have American coarse, square, Acme or modified Acme threads as specified herein.

3.3 Threaded parts. Threaded parts shall be in conformance with the applicable requirements of NBS H28.

3.4 Identification marking. Each clamp and hand screw shall be marked in a legible and permanent manner with the manufacturer's name or with a U. S. registered trademark of such known character that the source of manufacture may be readily determined.

3.5 Finish. All surfaces of the clamps and hand screws shall have a smooth finish with all edges rounded or beveled and shall be thoroughly cleaned and free of all foreign matter, fins, scale, flux, or any other imperfection.

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3.6 Type I, C clamp. The type I C clamp shall consist essentially of a metal frame and screw, assembled as an integral unit, and shall be capable of completing all tests specified herein. The working depth shall be the distance measured from the centerline of the screw to the maximum depth that a block having a dimension equal to the full opening dimension can be inserted into the clamp.

3.6.1 Frame. The anvil of the frame shall have a smooth flat surface perpendicular to the centerline of the class 2B, or better, right-hand threaded hole.

3.6.2 Screw. All type I clamps shall have a metal screw with a class 2A, or better, right-hand thread. All type I clamps, except the class B, style 2 and the class D shall have a nonremovable slip-bar type handle 2 inches long, minimum, inserted through the outer end of the screw. All clamps, except the class D, shall have a swivel button securely attached to the inner end of the screw. The swivel button shall be free to rotate and tilt on the screw when under no load but shall not rotate when under load by rotation of the screw. The face of the swivel button shall have a smooth, flat surface. See figure 3 and 3.6.6 for the description of the handle and the button on the class B, style 2 and the class D clamps.

3.6.3 Class E clamps. The type I, class E clamps shall be satisfactory for very light service use, be either style 1 or 2, and the size as specified (see 6.1). They shall be similar to figure 1 and conform to the applicable requirements specified in table I.

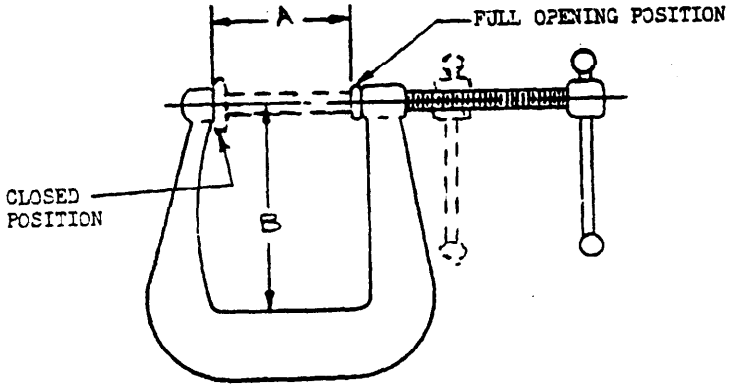


FIGURE 1. Type I, class E, very light service clamp.

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TABLE I. Type I, class E, very light service clamps

Nominal size of clamp	A Full opening (minimum)	B Working depth (minimum)	Diameter of screw (minimum)	Weight of clamp (maximum)	Test load (minimum)
Style 1 - Regular throat					
	Inches	Inches	Inch	Ounces	Pounds
1	1	7/8	1/4	4	500
1-1/4	1-1/4	1-1/8	1/4	5	800
1-1/2	1-1/2	1-1/2	5/16	6	850
2	2	1-1/8	5/16	9	875
2-1/2	2-1/2	2-1/2	3/8	16	900
3	3	1-1/4	3/8	12	1100
Style 2 - Deep throat					
1	1	1-3/8	1/4	5	500
2	1	3-1/2	1/4	6	200

3.6.4 Class A clamps. The type I, class A clamps shall be satisfactory for light service use, be either style 1 or 2, and the size specified (see 6.1). They shall be similar to figure 2 and shall conform to the applicable requirements specified in table II. The threaded portion of the screw shall be made of steel having a minimum tensile strength of 100,000 pounds per square inch (p.s.i.) and hardened to Rockwell C 22 minimum.

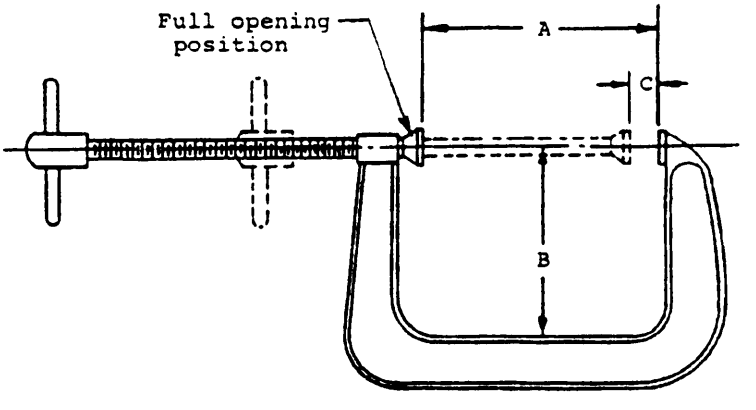


FIGURE 2. Type I, class A, light service clamp.

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TABLE II. Type I, class A, light service clamps

Nominal size of clamp	A Full opening (minimum)	C Closed opening (maximum)	B Working depth (minimum)		Diameter of screw, (minimum)	Weight of clamp (maximum)	Test load, (minimum)
			Style 1, regular throat	Style 2, deep throat			
	Inches	Inches	Inches	Inches	Inch	Pounds	Pounds
2	2	0	1¾	1 15/16	5/16	1-3/4	2700
3	3	0	1 7/8	2 5/16	½	2-1/2	3000
4	4	0	2¼	2 11/16	½	3-1/2	3700
6	6	0	2¾	3½	5/8	5	4500
8	8	2	3¼	4 3/8	5/8	7-3/4	5200
10	10	3	3½	5¼	¾	10-1/4	6000
12	12	4	3½	5 5/8	¾	14-3/4	8000

3.6.5 Class B clamps. The type I, class B clamps shall be satisfactory for medium service use, be either style 1 or 2, and the size specified (see 6.1). They shall be similar to figure 3 and conform to the applicable requirements specified in table III. The threaded portion of the screw shall be made of steel having a minimum tensile strength of 100,000 p.s.i. and be hardened to Rockwell C 22 minimum.

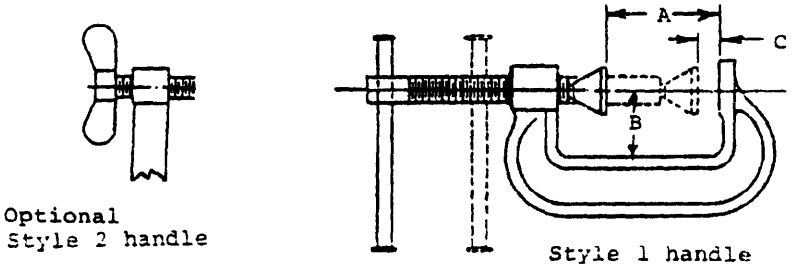


FIGURE 3. Type I, class B, medium service clamp.

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TABLE III. Type I, class B, medium service clamps

Nominal size of clamp	A Full opening (minimum)	C Closed opening (maximum)	B Working depth (minimum)	Diameter of screw, (minimum)	Dimensions of handle		Weight of clamp (maximum)	Test load (minimum)
					Diameter, (minimum)	Length, (minimum)		
			Style 1 - Regular throat					
	Inches	Inches	Inches	Inch	Inch	Inches	Pounds	Pounds
4	3 3/4	3/8	2 1/4	3/4	3/8	6 1/2	4 3/4	10,000
6	5 3/4	2 5/8	2 1/4	3/4	3/8	6 1/2	6	11,250
8	7 3/4	4 11/16	2 1/2	3/4	3/8	6 1/2	7 1/4	12,500
10	9 3/4	6 11/16	2 1/2	3/4	3/8	6 1/2	8 1/2	13,750
12	11 3/4	8 11/16	2 1/2	3/4	3/8	6 1/2	12	15,000
15	14 3/4	10	3	7/8	7/16	7 1/2	16	16,250
18	17 3/4	13	3	7/8	7/16	7 1/2	20	17,500
			Style 2 - Deep throat					
8	8	4-3/4	12	3/4	See figure 3		13	1,800

3.6.6 Class D clamps. The type I, class D clamps shall be satisfactory for heavy service use, be similar to figure 4, and conform to the applicable requirements specified in table IV for the size specified (see 6.1). The screw shall be made of steel having a minimum tensile strength of 125,000 p.s.i. and be hardened to Rockwell C 22 minimum. The outer end of the screw shall have square wrenching surface with an across the flats dimension not less than the major diameter of the screw. The inner end of the screw shall have a cylindrical portion with a diameter approximately equal to the minor diameter of the threaded portion of the screw.

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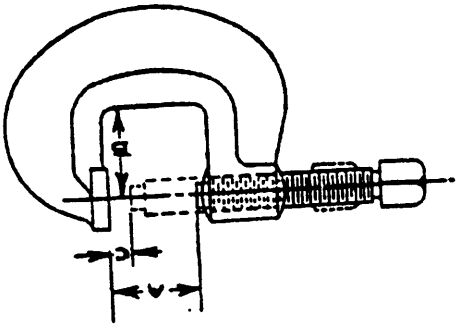


FIGURE 4. Type I, class D, heavy service clamp.

TABLE IV. Type I, class D, heavy service clamps

Nominal size of clamp	A Full opening (minimum)	C Closed opening (maximum)	B Working depth (minimum)	Diameter of screw (minimum)	Weight of clamp (maximum)	Test load (minimum)
	Inches	Inches	Inches	Inch	Pounds	Pounds
3/4	3/4	0	9/16	5/16	1/2	2,500
1	1	0	15/16	3/8	3/4	5,600
1-1/2	1-1/2	3/4	1-7/16	7/16	1-3/4	8,750
2	2	7/8	1-11/16	9/16	3-1/2	12,500
3	3	1-3/8	2-3/16	3/4	6-1/2	16,250
4	4	2	2-5/16	7/8	10	20,000
4-1/2	4-1/2	2	2-11/16	7/8	10	20,000
5	5	2-1/2	2-15/16	7/8	13-1/2	23,800
6	6	3-1/8	3	1	18-1/2	27,500
8	8	4-1/4	3-7/16	1-1/8	26-1/2	31,250
10	10	6-3/8	3-13/16	1-1/8	34	35,300
12	12	7-5/8	3-15/16	1-1/8	43-1/2	40,000

3.7 Type III, quick adjusting screw clamp. The heads and jaws of all type III clamps shall be either a steel forging or malleable or ductile iron. The face of the jaws and heads that contacts the item being clamped shall have a smooth flat surface and shall be approximately 90° to the centerline of the bar.

3.7.1 Type III, class A. The type III, class A clamp shall consist essentially of a bar, fixed head, screw, crank, an adjustable and a movable jaw. The clamps shall be similar to figure 5 and shall conform to the requirements specified herein and in table V for the size specified (see 6.1).

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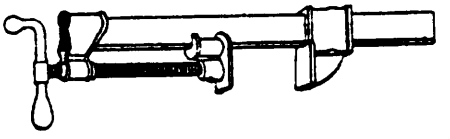


FIGURE 5. Type III, class A, clamp, screw quick-adjusting, fixed head with adjustable jaw.

TABLE V. Type III, class A, clamp, screw, quick-adjusting, fixed head with adjustable jaw

Nominal size of clamp	Capacity (distance between jaws), (minimum)	Diameter of screw, (minimum)	Width of faces of jaws		Weight of clamp	Test load,
			(Maximum)	(Minimum)	(maximum)	(minimum)
	Feet	Inch	Inches	Inches	Pounds	Pounds
1½	1½	5/8	2¼	1¾	10	6000
2	2	5/8	2¼	1¾	10½	6000
2½	2½	5/8	2¼	1¾	11½	6000
3	3	5/8	2¼	1¾	12½	6000
4	4	5/8	2¼	1¾	14½	6000
5	5	5/8	2¼	1¾	16½	6000
6	6	5/8	2¼	1¾	18½	6000
7	7	¾	2¼	1¾	24	9000
8	8	¾	2½	1¾	27	9000
9	9	¾	2½	1¾	29	9000
10	10	¾	2½	1¾	31	9000

3.7.1.1 Bar. The bar shall be of steel and shall be I-beam or rectangular shaped in cross section with the width and thickness uniform for the entire length. At the end opposite the fixed head, the bar shall be provided with a device which will prevent the unintentional removal of the movable jaw from the bar.

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3.7.1.2 Fixed head. The fixed head shall be securely attached in a rigid manner to one end of the bar. There shall be a hole in the fixed head, the axis of which is substantially parallel to the axis of the bar, and not less than 7/8 inch from the nearest edge of the bar. The hole shall have right-hand, square, or Acme threads to receive the screw.

3.7.1.3 Screw. The screw shall be of steel possessing a minimum tensile strength of 60,000 p.s.i. and be hardened to Rockwell C 22 minimum. The threads shall be right-hand, square, or Acme threads which shall provide a class 2 fit, or better, with the threaded hole in the fixed head. The screw shall have a travel of not more than 6 inches.

3.7.1.4 Crank. The crank shall be a balanced "T" type, of one-piece construction, either a steel forging, or malleable or ductile iron, and shall be secured to the outer end of the screw.

3.7.1.5 Adjustable jaw. The adjustable jaw shall have an opening to receive the bar, and shall be easily movable along the bar. The adjustable jaw shall be securely attached to the inner end of the screw so that under no load the screw will rotate easily. A hardened steel plate or other device shall be provided to prevent excessive wear where the end of the screw bears on the adjustable jaw.

3.7.1.6 Movable jaw. The movable jaw shall have an opening to receive the bar, and when not locked shall be easily movable along the bar. The jaw shall be provided with a positive mechanical locking device which, when locked at any desired position along the bar, shall prevent longitudinal motion of the jaw.

3.7.2 Type III, class B. The type III, class B clamp shall consist essentially of a bar, sliding head, screw, and a fixed jaw. The clamps shall be similar to figure 6 and conform to the requirements specified herein and in table VI for the size specified (see 6.1).

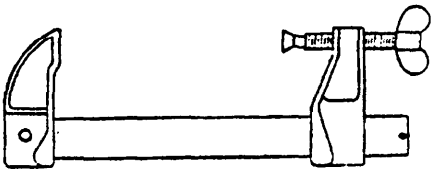


FIGURE 6. Type III, class B clamp, screw, quick-adjusting, sliding head.

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TABLE VI. Type III, class B clamp, screw, quick-adjusting, sliding head

Nominal size of clamp	Capacity (distance between swivel button and jaw), (minimum)	Distance from axis of screw to edge of bar, ±10 percent	Diameter of screw, (minimum)	Weight of clamp (maximum)	Test load, (minimum)
	Inches	Inches	Inch	Pounds	Pounds
4	4	2	½	3	1800
5	5	2	½	4	1800
6	6	2	½	4½	1800
8	8	2	½	5	1800
10	10	2¾	9/16	7½	2400
12	12	2¾	9/16	8	2400
18	18	2¾	⅝	9	4500
24	24	2¾	⅝	10¼	4500
30	30	2¾	⅝	11½	4500
36	36	2¾	⅝	12½	4500
48	48	2¾	⅝	14½	4500

3.7.2.1 Bar. The bar shall be of steel and the width and thickness shall be uniform for the entire length. At the end opposite the fixed jaw, a device shall be securely attached to the bar which shall prevent the unintentional removal of the sliding head from the bar.

3.7.2.2 Sliding head. The sliding head shall have an opening to receive the bar, and when not locked shall be easily movable along the bar. The sliding head shall be provided with a locking device which at any desired position along the bar shall prevent longitudinal motion of the head. There shall be a hole in the sliding head, the axis of which is parallel to the axis of the bar. The hole shall have right-hand threads.

3.7.2.3 Screw. The screw shall be of steel possessing a minimum tensile strength of 60,000 p.s.i. and hardened to Rockwell C 22 minimum. The threads shall be right-hand, and shall provide a class 2 fit, or better, with the threaded hole of the sliding head. The outer end of the screw shall be provided with a nonremovable handle. The handle may be integral with the screw, or a separate piece securely attached to the screw. A swivel button having a spherical seat shall be securely attached to the inner end of the screw so that under no load the button will be free to rotate and tilt easily on the screw. The face of the swivel button shall have a smooth, flat surface, but shall not rotate when a load is applied to the clamp and the screw rotated. The screw shall have a travel of not more than 4 inches.

3.7.2.4 Fixed jaw. The fixed jaw shall be securely attached to one end of the bar.

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3.7.3 Type III, class C. The type III, class C clamp shall consist essentially of a threaded head, screw, crank-type handle, an adjustable and a movable jaw. The clamp shall be similar to figure 7 and conform to the requirements specified herein. The threaded pipe bar shown in figure 7 (supporting the jaws) is not furnished with the clamp (see 4.4.1.2).

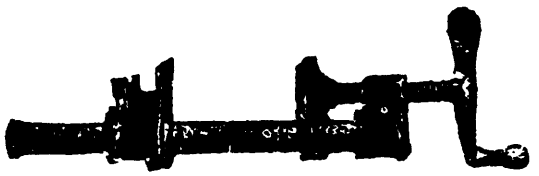


FIGURE 7. Type III, class C, clamp, screw, quick-adjusting, threaded head.

3.7.3.1 Threaded head. The threaded head shall contain two holes which are substantially parallel to each other. One hole shall have a 5/8-inch, right-hand, square or Acme thread through its entire length to receive the screw. The other hole shall be counterbored and threaded to receive a 3/4-inch iron pipe. The counterbore shall have a minimum diameter of 1.060 inches for approximately one-half the depth of the hole and the remainder of the hole shall have 3/4-14 NPT threads.

3.7.3.2 Screw. The screw shall be of steel possessing a minimum tensile strength of 75,000 p.s.i. and hardened to Rockwell C 24 minimum. The threads shall provide a class 2 fit, or better, with the threaded head. The screw shall provide a travel of not less than 1-1/2 inches to the adjustable jaw.

3.7.3.3 Crank-type handle. The crank-type handle shall be of one-piece construction, steel, malleable or ductile iron, and shall be securely attached to the outer end of the screw.

3.7.3.4 Adjustable jaw. The adjustable jaw shall have a hole to receive the outside diameter of a 3/4-inch iron pipe with sufficient clearance on the pipe to insure easy movement along the pipe. The face of the jaw shall be approximately 1-3/4 by 1-3/4 inches in size. The adjustable jaw shall be securely attached to the inner end of the screw in such a manner that under no load the screw will rotate easily. A hardened steel plate or other device shall be provided to prevent excessive wear where the end of the screw bears on the adjustable jaw.

3.7.3.5 Movable jaw. The movable jaw shall have a 1.060-inch minimum diameter opening to receive a 3/4-inch iron pipe, which when the locking device is released, shall allow easy movement along the pipe. The jaw shall be provided with a positive mechanical locking device which when locked shall prevent longitudinal movement of the jaw along the pipe when subjected to a 2800-pound test load, as specified in 4.4.1. The face of the jaw shall be approximately 1-3/4 by 1-3/4 inches in size.

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3.7.4 Type IV, hand screw. The type IV hand screw shall consist essentially of two jaws of the same nominal dimensions, two screws with two handles, and shall be designed to permit the jaw to grip angular or straight work, or so that one jaw may be made to overlap the other. The hand screw shall be similar to figure 8 and conform to the requirements specified herein and in table VII for the size specified (see 6.1).

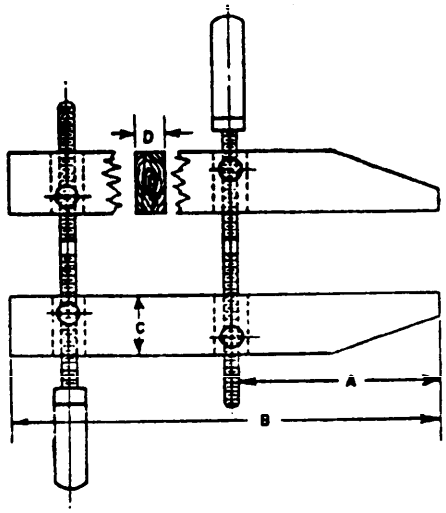


FIGURE 8. Type IV, hand screw, adjustable.

TABLE VII. Type IV, hand screw, adjustable

Nominal size of hand screw	Opening between jaws, (minimum)	A Distance from end of jaw to middle screw ±1/4	Diameter of screws, (minimum)	Dimensions of jaws			Weight of hand screw complete, (maximum)
				B ±1/4	C ±5 percent	D ±5 percent	
	Inches	Inches	Inch	Inches	Inches	Inches	Pounds
2	2	2	1/4	4	1 1/16	1 1/32	5/8
3	3	3	5/16	6	1 3/8	1 1/4	1-1/4
4 1/2	4 1/2	4	3/8	8	1 3/4	1 1/2	2-1/4
6	6	5	3/8	10	1 15/16	1 5/8	2-3/4
8 1/2	8 1/2	6	7/16	12	2 1/4	1-7/8	4-3/4
10	10	7	7/16	14	2 3/8	2 1/16	5-3/4
12	12	8	1/2	16	2 5/8	2 3/16	7-3/4

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3.7.4.1 Jaws. Each jaw shall be made of hard maple seasoned in a manner to preclude warping and shall be smoothly finished. Two cylindrical steel bearings shall be assembled in holes provided in each jaw. Each bearing shall have a transverse hole threaded to receive the spindle. The holes in the jaw through which the spindle screws pass shall be slotted longitudinally. Each jaw shall be dipped in oil or coated with a material which will prevent the absorption of moisture. The wood shall be seasoned to show not more than 12 percent moisture content (see 4.4.2).

3.7.4.2 Screws. Each screw shall be of steel and shall have right-hand threads at one end and left-hand threads at the other. The threads shall provide a class 2 fit, or better, with the threaded holes in the bearings. The right-hand rotation of the screws shall cause the faces of the jaws to come into contact. One end of each screw shall be provided with a handle securely attached to the screw.

3.7.4.3 Handles. The handles shall be of such size and shape as will provide a comfortable and adequate grip for the hand. Each handle shall be of good quality, straight-grained hardwood such as mahogany, beech, maple, birch, cherry, or goncola alves (astronium), and shall be provided with a substantial metal ferrule securely attached to the handle at the end into which the screw is inserted. Each handle shall be smoothly finished and provided with a protective coating which shall prevent the absorption of moisture.

3.7.5 Type V, clamp, cantilever jaw. The type V cantilever jaw clamp shall be of steel and consist essentially of a body, movable jaw, pressure pads and screw, be similar to figure 9 and conform to the requirements specified herein and in table VIII for the size specified (see 6.1).

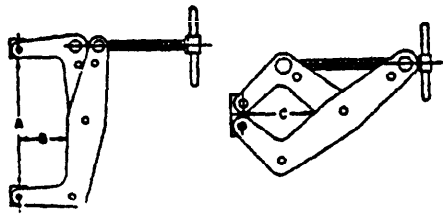


FIGURE 9. Type V, clamp cantilever jaw.

TABLE VIII. Type V, clamp, cantilever jaw

Nominal size of clamp	A Full opening (minimum)	Working depth of throat		Diameter of screw, (minimum)	Weight of clamp complete, (maximum)	Test load, (minimum)
		B Open (minimum)	C Closed (minimum)			
	Inches	Inches	Inches	Inch	Pounds	Pounds
2	2	1⅛	2	¼	½	950
3	3	1¼	2¾	⅜	1¼	1700
6	6	2	3¼	½	3	2700
12	12	4½	6⅞	¾	13½	7500

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3.7.5.1 Body. The body shall consist of two steel side plates separated by shoulder rivets and shall have a square pressure pad retained on one end in such a manner as to oppose the similar pressure pad on the movable jaw. A threaded trunnion shall be fitted on the end of the body opposite the pressure pad to receive the screw.

3.7.5.2 Movable jaw. The movable jaw shall consist of two steel side plates separated by shoulder rivets and shall have a square pressure pad retained by a rivet at one end. A free rotating thrust trunnion shall be mounted in the jaw so as to receive the end of the screw.

3.7.5.3 Pressure pads. The pressure pads shall be made of square, unhardened, mild steel with three smooth surfaces and one V-groove face. The pressure pads shall rotate freely on their retaining rivets.

3.7.5.4 Screw. The screw shall be of steel, heat treated to resist wear and shall have right-hand threads which shall provide a class 2 fit, or better, with the threaded trunnion of the body. Screws for clamp sizes 2, 3, and 6 inches shall have national coarse threads. The screw for the 12-inch size clamp shall have Acme threads. The outer end of the screw shall be fitted with a nonremovable handle except the 12-inch size which shall be square and provided with a hole of adequate size for insertion of a round pin for tightening purposes.

3.8 Workmanship. The clamps and hand screws shall be free of any defects that can adversely affect their operation, serviceability, or life expectancy.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling for examinations and tests.

4.2.1 Inspection lot. An inspection lot shall consist of all clamps or hand screws of one type, class, style, and size offered for acceptance at one time. The sample unit shall be one clamp or hand screw.

4.2.2 Sampling for examinations and tests. Sampling for all examinations and tests shall be accomplished in accordance with MIL-STD-105 at inspection level II with an Acceptable Quality Level (AQL) of 2.5.

4.3 Examinations.

4.3.1 Each sample selected in accordance with 4.2.2 shall be examined for conformance to the requirements for dimensions, finish, markings, and workmanship. Any sample that does not comply with all requirements shall be classified as a reject.

4.3.2 Examination of preparation for delivery. Examination shall be made to determine compliance with preservation, packaging, packing, and marking requirements in accordance with PPP-P-40.

4.4 Tests. Each sample clamp and hand screw shall be tested in accordance with 4.4.1 through 4.4.4, as applicable.

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4.4.1 Proof load test for types I, III, and V clamps shall be conducted on a tension machine or similar device of known accuracy. Each sample clamp shall be subjected to the applicable specified test load. The test shall be conducted with the clamp in the full open or minimum capacity opening as applicable. The test load shall be applied gradually, to simulate actual usage, until the specified minimum load is obtained. This load should be maintained for a period of not less than 5 minutes. Upon completion of this test, the clamp shall be examined and any evidence of deformed or damaged components shall be sufficient cause for classifying that clamp as a reject. The iron pipe used with the type III, class C clamp (see 4.4.1.2) is excluded from this examination.

4.4.1.1 Type I clamps. The test load for the type I clamps shall be applied as close as practical to the center line of the screw.

4.4.1.2 Type III clamps. The test load for the type III clamps shall be applied as close as practical to the centerline of the screw and parallel to the center line of the bar. The class C clamp shall be assembled on a 30-inch length of standard 3/4-inch threaded iron pipe for this test with the face of the jaws fixed in a position 24 inches apart.

4.4.2 Type IV hand screw. The manufacturer shall be required to furnish certification that the jaws of the type IV hand screws are made of material as specified in 3.7.4.1. The jaws shall be subjected to a moisture content test utilizing an electrical moisture meter to determine conformance with 3.7.4.1. The moisture content shall be determined in accordance with instructions furnished by the manufacturer of the electrical moisture meter. The electrical moisture meter shall not be used on jaws that are surface-wet or that have been color-stained. In determining the moisture content of the jaws, the reading on the electrical moisture meter shall be taken when the meter electrodes are driven to the center of the jaw or handle being tested.

4.4.3 Tensile test. The manufacturer shall be required to furnish certification that the material of the screws conform to minimum tensile strength specified.

4.4.4 Hardness test. The sample type I clamps shall be tested for the hardness specified for the screws. Test impressions may be taken at any location near the extreme end of the threaded portion.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The clamps or hand screws shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. Preservation and packaging shall be level A, B, or C as specified (see 6.1), packing shall be level A, B, or C as specified (see 6.1), and marking shall be for civil agencies or military activities as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and size required (see 1.2.1).
(c) Level of preservation, packaging, packing, and type of marking required (see 5.1).

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6.2 Supersession data.

6.2.1 The grades a and b, type I clamps specified in GGG-C-406C have been combined and the grade designations have been deleted from GGG-C-406D. This change was effected since the requirements for these clamps are based upon test loads in lieu of specific materials and methods of forming. The test loads of the grade a clamps were used in GGG-C-406D. The style 3 has been replaced by the style 1.

6.2.2 The following clamps shown in GGG-C-406C were not included in GGG-C-406D since no Federal Stock Numbers were assigned to these clamps and this indicates there are no requirements for them:

Type I, styles 4 and 5.
Type II, classes A and B.
Type III, class A, style 2.
Type III, class B, all sizes with a "distance from axis of screw to edge of bar greater than 2-3/4 inches".

6.2.3 The type VI clamps covered in GGG-C-00406b are covered in GGG-W-00649.

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 40 cents each.

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